16. CSS Styling in React

▼ What is CSS Styling in React?

- **CSS Styling in React** means applying visual styles (colors, layout, font, spacing, etc.) to your React components to make them look better.
- React doesn't restrict how you style components - you can use traditional CSS, inline styles, CSS-in-JS, utility frameworks, or component-based styling.
- React encourages **component-based styling**, where styles are scoped and reusable per component, avoiding global conflicts.

▼ Different Ways to Style in React

- We can perform CSS styling in many ways in React as follows:
  1. Inline Styling
  2. External Stylesheet (Traditional Way)
  3. CSS Modules (.module.css)
  4. Styled Components (CSS-in-JS)
  5. By Using Tailwind CSS
  6. By Using Bootstrap

▼ 1. Inline Styling

- Inline styling is when styles are applied **directly** to an element using the `style` attribute.
- In React, the value must be a `JavaScript object` with `camelCase` property names.
- By using inline styles you have some limitations - you can't use pseudo-selectors, media queries, keyframes, etc.
- **Syntax:**

```
style = {{key: "value"}}
style = {{fontSize: "20px"}}
```

- **For Example →**
- **Code:**

```
import React from "react";

const InlineStyle = () => {
  return (
    <div>
      <h1 style={{ color: "red", fontSize: "25px" }}>Inline Style</h1>
      <p style={{ color: "green" }}>
        Inline styling is when styles are applied directly to an element using
        the style attribute.
      </p>
    </div>
  );
};

export default InlineStyle;
```

- **Output:**

Inline Style

Inline styling is when styles are applied directly to an element using the style attribute.

- ✅**Use When:**
  - You need quick, simple styling.
  - You want to apply **dynamic** or conditional styles (based on state or props).
- ❌**Avoid When:**
  - Makes JSX messy if styles are large.

- Styling complex components.
- You need media queries or pseudo-classes like `:hover`, `:focus`, etc.

▼ 2. External Stylesheet

- A separate `.css` file that contains global styles, imported into your component.
- **For Example →**
- **Code: Component1.jsx**

```
import React from "react";
import "./style1.css";

const Component1 = () => {
  return (
    <div>
      <h1>Component1</h1>
      <button className="button">Click Me</button>
    </div>
  );
};

export default Component1;
```

- **Code: style1.css**

```
.button {
  background-color: blue;
  color: white;
  padding: 10px 20px;
  border-radius: 8px;
  border: none;
  cursor: pointer;
}
```

- **Output:**

Component1

Click Me

- **Drawback:** Styles are **global**, not scoped per component.
- **Code: Component2.jsx**

```
import React from "react";
import "./style2.css";

const Component2 = () => {
  return (
    <div>
      <h1>Component2</h1>
      <button className="button">Click Me</button>
    </div>
  );
};

export default Component2;
```

- **Code: style2.css**

```
.button {
  background-color: green;
  color: white;
  padding: 10px 20px;
  border-radius: 8px;
  border: none;
  cursor: pointer;
}
```

- **Output:**

Component1

Click Me

Component2

Click Me

- Here, if you observe here for both components button is green colored. But I have given a separate color, still, it is applying the same color.
- This is because whoever import statement that is nearest to the component to which CSS will be applied to the component button.
- **Code: App.jsx**

```
import React from "react";
import Component1 from "./CSS Styling/Component1"; // Button - Blue
import Component2 from "./CSS Styling/Component2"; // Button - Gree
n

const App = () => {
  return (
    <div>
      <Component1 />
      <Component2 />
    </div>
  );
};

export default App;
```

- Here, `Component2` is nearest to the components and color of the button is **green**, that why it is applied to both buttons.

▼ 3. CSS Modules (`.module.css`)

- A local-scoped CSS file where class names are **unique to the component** using it.
- Prevents global name conflicts.
- **For Example →**
- **Code: Component1.jsx**

```react
import React from "react";
import Styles from "./Style1.module.css";

const Component1 = () => {
  return (
    <div>
      <h1>Component1</h1>
      <button className={Styles.button}>Click Me</button>
    </div>
  );
};

export default Component1;
```

- **Code: Style1.module.css**

```css
.button {
  background-color: blue;
  color: white;
  padding: 10px 20px;
  border-radius: 8px;
  border: none;
  cursor: pointer;
}
```

- **Code: Component2.jsx**

```react
import React from "react";
import Styles from "./Style2.module.css";

const Component2 = () => {
```

```
  return (
    <div>
      <h1>Component2</h1>
      <button className={Styles.button}>Click Me</button>
    </div>
  );
};

export default Component2;
```

- **Code: Style2.module.css**

```
.button {
  background-color: red;
  color: white;
  padding: 10px 20px;
  border-radius: 8px;
  border: none;
  cursor: pointer;
}
```

- **Output:**

▼ 4. Styled Components (`CSS-in-JS`)

- Styled-Components is a library for React that allows you to write your CSS directly in your JavaScript.

- This is called “ `CSS-in-JS` ”.
- A library that lets you write **actual CSS** in JavaScript using tagged template literals.
- Styles are scoped to the component and can use props/state.
- **Command to install the library:**

💡 npm install styled-components

- **Import:**

💡 Import → import Style from “styled-components”

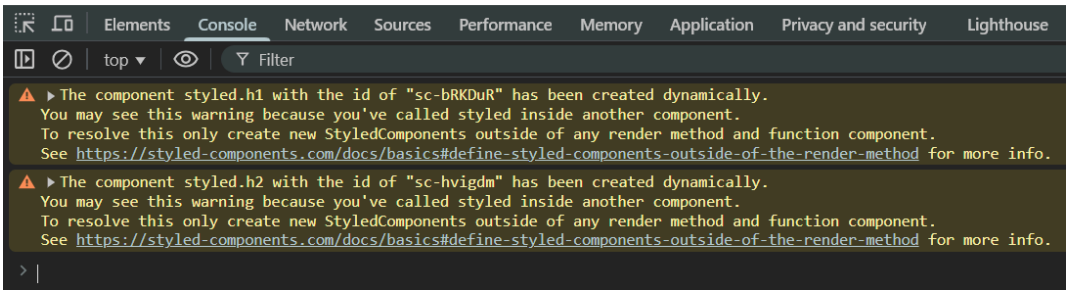
- **For Example →**
- **Code:**

```
import React from "react";
import styled from "styled-components";

const StyledComponent = () => {
  let H1 = styled.h1`
    color: red;
    font-size: 20px;
  `;
  let H2 = styled.h2`
    color: blue;
    font-size: 18px;
  `;
  return (
    <>
      <H1>Hello h1 style coming from styled components</H1>
      <H2>Hello h2 style coming from styled components</H2>
    </>
  );
};
```

```
export default StyledComponent;
```

- **Output:**

Hello h1 style coming from styled components
Hello h2 style coming from styled components



- **Supports hover, media queries, nesting, and dynamic props:**
- **Code:**

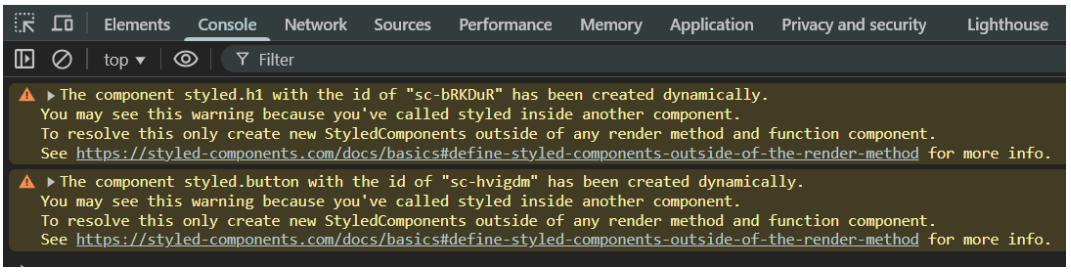
```
import React from "react";
import styled from "styled-components";

const StyledComponent = () => {
  let H1 = styled.h1`
    color: red;
    font-size: 20px;
  `;

  let Button = styled.button`
    padding: 8px 10px;
    background-color: blue;
    cursor: pointer;
    color: white;
    border-radius: 5px;
    margin: 10px;
    &: hover {
      background-color: red;
    }
```

```
  `;

  return (
    <>
      <H1>Hello h1 style coming from styled components</H1>
      <Button>Click Me</Button>
    </>
  );
};

export default StyledComponent;
```

- **Output:**

Hello h1 style coming from styled components

Click Me



▼ 5. By Using Tailwind CSS

- A utility-first CSS framework where you style elements by composing **predefined class names.**

▼ Installation Process of Tailwind CSS:

1. **Step-1:** Go to this website - https://tailwindcss.com/docs/installation/using-vite
2. **Step-2:** Copy the first command and open new terminal - Command prompt – and paste the command and hit enter.

01 **Install Tailwind CSS**

Install `tailwindcss` and `@tailwindcss/vite` via npm.

Terminal

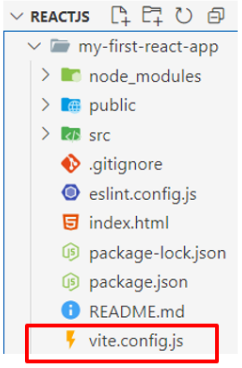
```
npm install tailwindcss @tailwindcss/vite
```

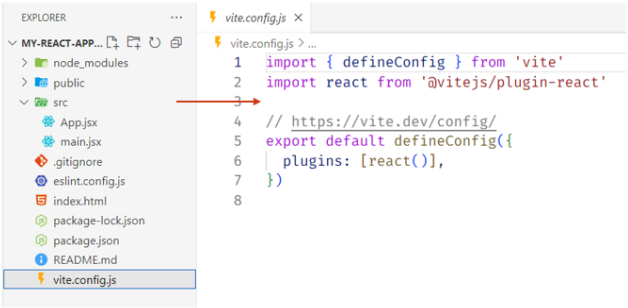
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Shridhan\Desktop\demo\my-react-app-with-tailwind>npm install tailwindcss @tailwindcss/vite
```

3. **Step-3:**

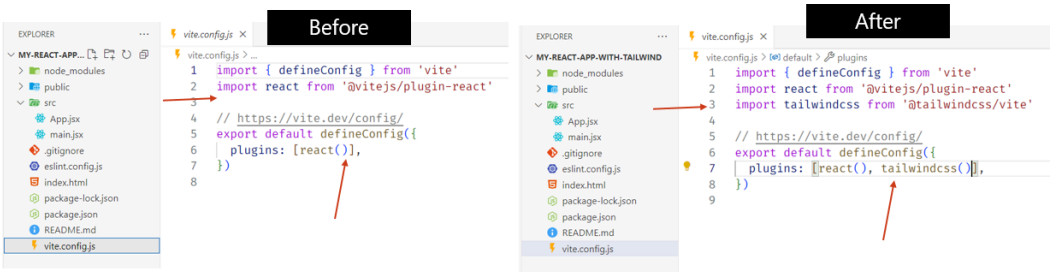
- Copy the 2nd import statement only - **import tailwindcss from '@tailwindcss/vite'**
- Then go to vs code you will see file **vite.config.js** at the last of the project.
- Open the file paste that import statement in the line no-3.





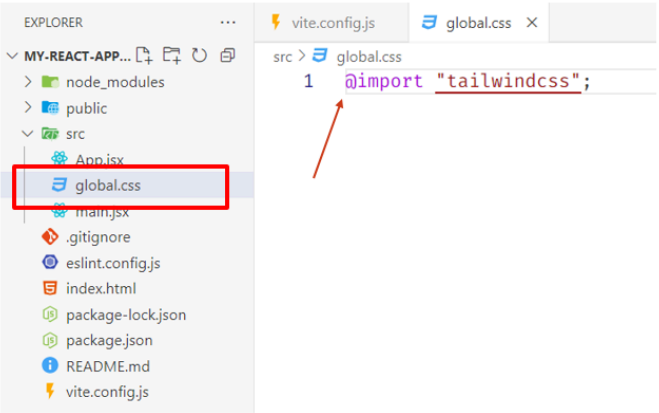
4. **Step-4:**

- Now copy the **tailwindcss()** from the plugins option.
- After that go to vs code, **vite.config.js** – inside you can see already react is their provide **one comma** (,) and paste that **tailwindcss()** their itself.



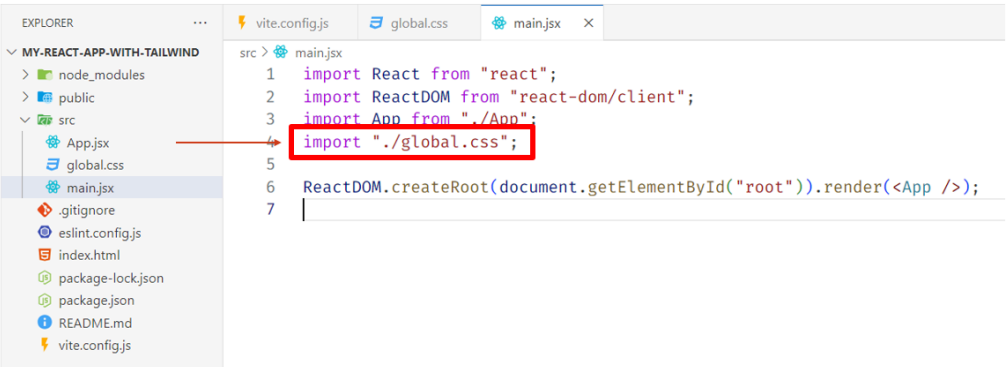
5. **Step-5:**

- Now create one file inside the **src** folder. – filename - **global.css**
- Inside that paste the 3rd command - **@import "tailwindcss";**



6. **Step-6:**

- Now import that global.css file import inside the **main.jsx** file.
- Or you can type exact same import statement inside your code in line-4



7. **Step-7:**

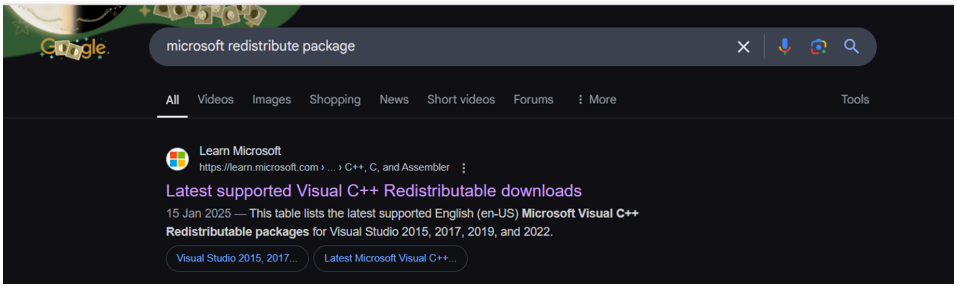
- Tailwind installation done for your system.

- Now just run the react project in terminal → command prompt → **npm run dev**
- Open the link in the browser that it done.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

32 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\Shridhan\Desktop\demo\my-react-app-with-tailwind>npm run dev

> my-react-app-with-tailwind@0.0.0 dev
> vite

Port 5173 is in use, trying another one...

  VITE v6.2.2  ready in 856 ms

  → Local:   http://localhost:5174/
  → Network: use --host to expose
  → press h + enter to show help
```

▼ For those who are getting errors

- If your are getting error like **lightningcss module not found.**
- Type **Microsoft redistribute package** in any browser and click on the first link.

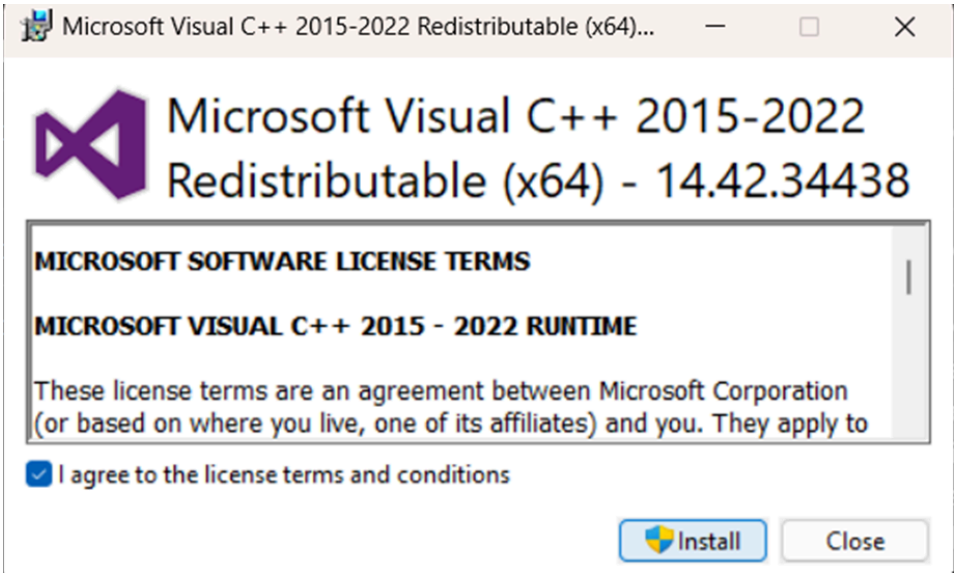


- Next thing just scroll the page you will see this type of one box.
- In the terminal, just check what it the version error coming and it may be asking like **x64** or **x86** according that just click on the link and download that package in your system.

Expand table

Architecture	Link	Notes
ARM64	https://aka.ms/vs/17/release/vc_redist.arm64.exe	Permalink for latest supported ARM64 version
X86	https://aka.ms/vs/17/release/vc_redist.x86.exe	Permalink for latest supported x86 version
X64	https://aka.ms/vs/17/release/vc_redist.x64.exe	Permalink for latest supported x64 version. The X64 Redistributable package contains both ARM64 and X64 binaries. This package makes it easy to install required Visual C++ ARM64 binaries when the X64 Redistributable is installed on an ARM64 device.

- After downloading just check the **I agree** and click on the **install** button.
- After the completion of installation just **restart your laptop**.
- Open the vs code and run the react project now that error will not come.



- **For Example →**
- **Code:**

```
import React from "react";

const TailwindCSSComponent = () => {
  return (
    <div>
```

```
      <h1 className="text-blue-600 text-3xl font-bold text-center p-5">
        Tailwind CSS
      </h1>
      <p className="text-teal-600 font-semibold text-center">
        A utility-first CSS framework where you style elements by composi
ng
        predefined class names.
      </p>
      <div className="w-[100%] flex justify-center mt-4">
        <button className="px-6 py-2 bg-teal-600 hover:bg-teal-700 text
-white rounded-md cursor-pointer">
          Click for More
        </button>
      </div>
    </div>
  );
};

export default TailwindCSSComponent;
```

- **Output:**